

EXPERIMENT 14: PERFORM TRANSFORMATION USING HOMOGRAPHY MATRIX

Aim

To perform image transformation using a Homography Matrix with Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Select four source and destination points.
4. Compute the homography matrix using `cv2.findHomography()`.
5. Apply `cv2.warpPerspective()`.
6. Display and save the transformed image.

Python Code

```
import cv2
import numpy as np

# Read input image
img = cv2.imread("input.jpg")

if img is None:
    print("Error: Image not found")
    exit()

height, width = img.shape[:2]

# Source points
src_pts = np.float32([
    [50, 50],
    [450, 50],
    [450, 300],
    [50, 300]
])
```

```
# Destination points
dst_pts = np.float32([
    [80, 80],
    [420, 40],
    [460, 310],
    [40, 280]
])

# Compute Homography Matrix
H, status = cv2.findHomography(src_pts, dst_pts)

# Apply Homography Transformation
output = cv2.warpPerspective(img, H, (width, height))

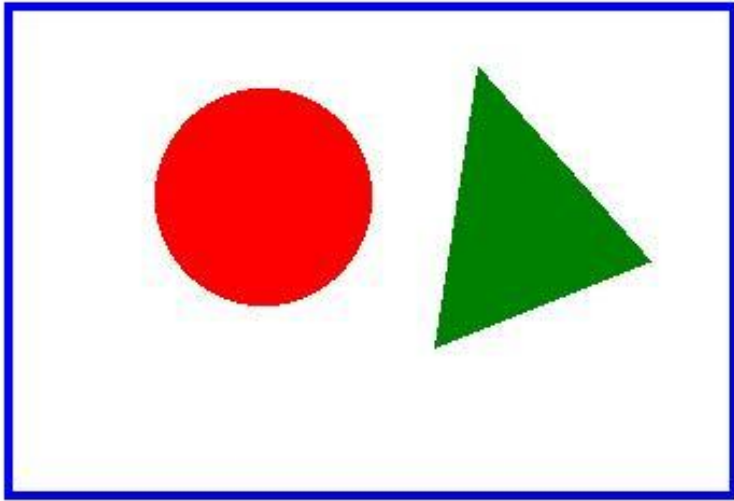
cv2.imwrite("homography_output.jpg", output)

cv2.imshow("Input Image", img)
cv2.imshow("Homography Output", output)

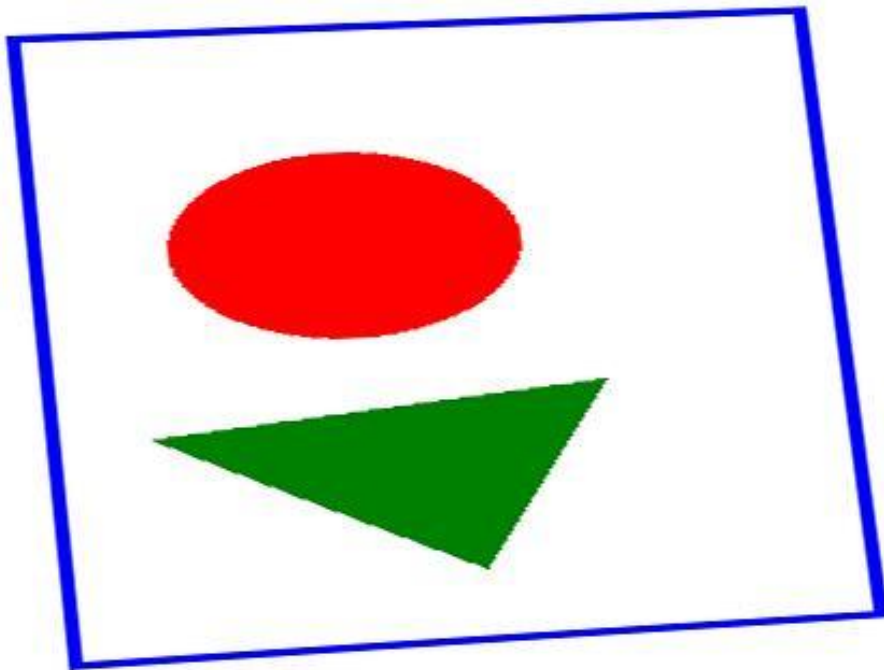
print("Homography Matrix:")
print(H)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the transformation using the Homography Matrix was successfully performed and the transformed image was obtained.